

# AISSAOUI Ismail

State Engineer in Artificial Intelligence & Data Science

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## PROFESSIONAL SUMMARY

State Engineer graduated from ESI-SBA, specializing in **Graph Neural Networks (GNN)**, Federated Learning, and the cybersecurity of Cyber-Physical Systems (CPS). Expert in structural modeling and centrality measures for distributed intrusion detection.

## EDUCATION

**National School of Computer Science (ESI-SBA, Algeria)** 2021 – 2026  
State Engineer's Degree & MSc in Artificial Intelligence

## RESEARCH ENGINEERING PROJECTS

**Career Connect — GNN recruitment Intelligence** 2024 – 2025  
Architected a **Graph Neural Networks** system for complex semantic matching.

- Modeled heterogeneous relationships using graph embeddings.
- Utilized centrality measures to identify pivotal nodes in dense networks. *Direct Match: Expertise in GNN and complex network theory.*

**Generative Digital Twin — Sonatrach** 2026  
Distributed inference and anomaly detection for complex industrial systems.

- Implemented **Mamba-SSM** and **xLSTM** for real-time monitoring.
- Secured Edge-Cloud data flow using Physics-Informed Neural Networks (PINN).

**University Scheduling (NP-Complete Optimization)** 2025  
Automated resource allocation using hybrid metaheuristics (**PSO/ACO/GA**).  
**Aura — Adaptive Reinforcement Learning System** 2025  
Trading platform with distributed real-time data pipelines and DQN agents.  
**Geo-RAG — Geospatial Knowledge Retrieval** 2025  
Semantic reasoning and RAG system for multi-source geospatial data.

## WORK EXPERIENCE

**AI Engineer Intern** Jan 2026 – Present  
**Sonatrach (Pipeline Transport Branch)**  
Predictive maintenance and software architecture for Digital Twins.  
**Network Engineering Intern** Sep 2024  
**Algérie Télécom RMC Center**

## TECHNICAL SKILLS

- Graph AI:** GNN (GraphSAGE, GAT), Centrality, PyTorch Geometric, NetworkX.
- Distributed AI:** Federated Learning (architecture), Privacy-preserving AI.
- Deep Learning:** PyTorch, Mamba-SSM, Transformers, PINN, RL.
- Security & Dev:** Intrusion Detection Systems (IDS), C++, Python, Docker, MLflow.

## LANGUES

Arabic (Native), English (Advanced), French (Fluent).